

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

Code of Conduct:

I Nithishkumaran M 192320037(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

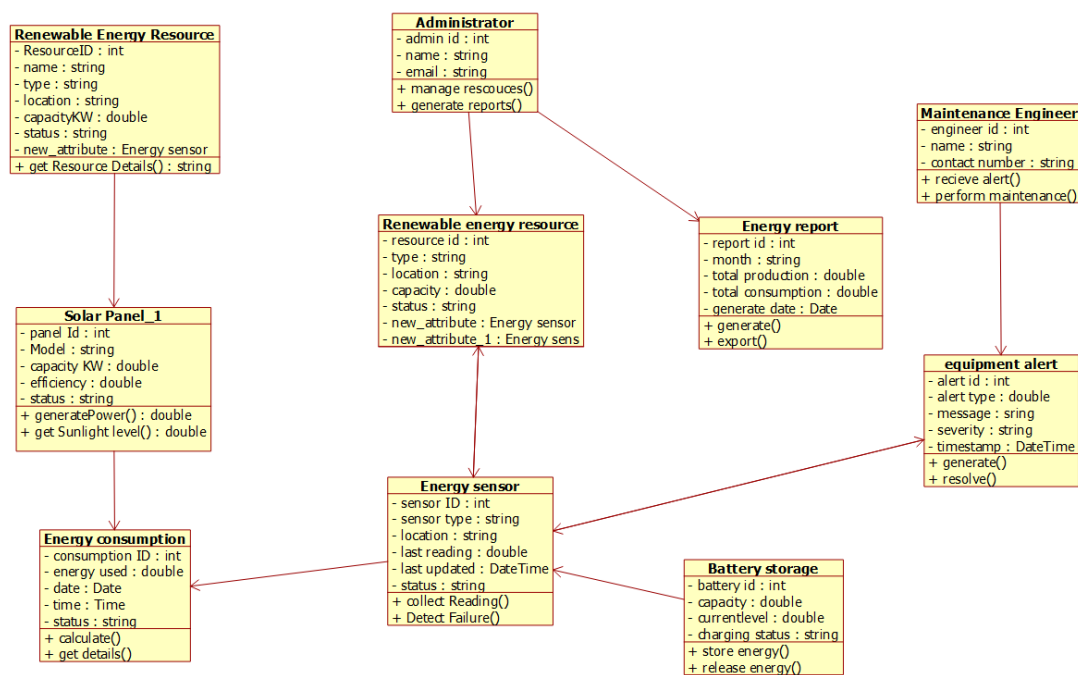
Nithishkumaran M

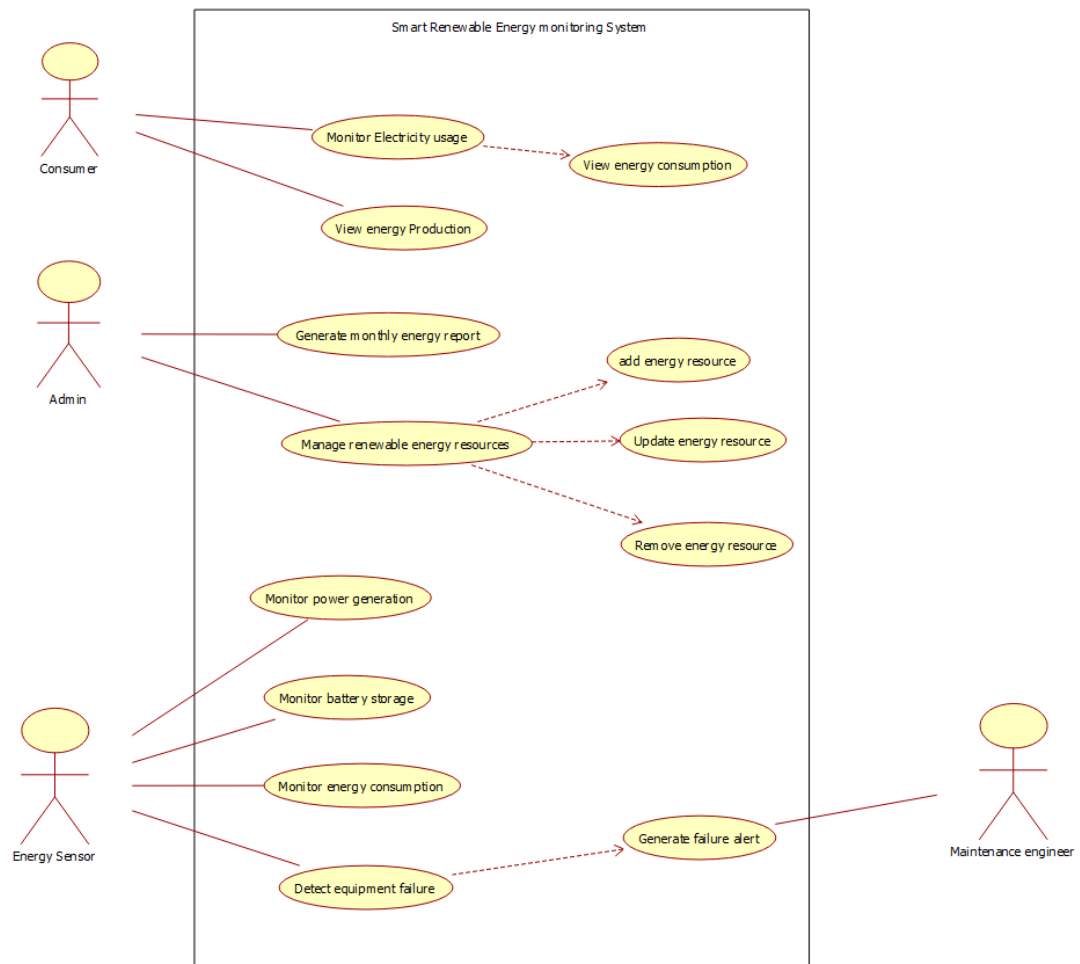
4.Smart Renewable Energy Monitoring System (SDG 7: Affordable and Clean Energy)

A smart city uses solar panels and wind turbines to generate electricity. Energy sensors monitor power generation, battery storage, and energy consumption. Consumers can monitor their electricity usage through a mobile application. Maintenance engineers receive alerts whenever equipment failures occur. The administrator generates monthly energy production reports and manages renewable energy resources.

Analyze the system and perform the following:

1. Draw the **Use Case Diagram**.
2. Identify all major **classes and attributes** and draw the **Class Diagram**.
3. Show suitable **relationships** among the classes using UML notations





RUBRICS FOR CALCULATION:**Case study**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand